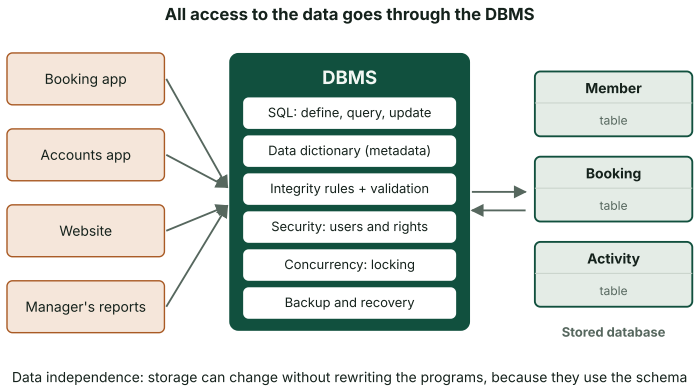
Data management: flat files, relational databases and the DBMS

IS01 FACT FILE

Specification: students should be able to

- Compare flat-file and relational approaches to storing data
- Describe the functions of a database management system (DBMS)
- Explain data independence, data redundancy and data inconsistency

Key words: flat file, relational database, DBMS, data redundancy, data inconsistency, data independence, data integrity, schema, data dictionary, concurrent access, program-data dependence



Application programs reach the data only through the DBMS, which provides SQL, the data dictionary, integrity, security, concurrency control and backup; this gives data independence

Why flat files cause problems

In a flat-file or traditional file-based system the same data is often held in several records or files. This data redundancy wastes storage and, more importantly, leads to data inconsistency: when one copy is updated and another is not, the organisation no longer knows which value is correct. Programs are also tied to the structure of their files (program-data dependence), so a small change to the file can force programs to be rewritten. Flat files remain acceptable for small, single-user lists.

The DBMS

A database management system is software that sits between users or application programs and the stored data. It provides a query language (usually SQL) for defining, updating and retrieving data, keeps a data dictionary describing every table and field, enforces validation and referential integrity, controls access through user accounts and permissions, manages concurrent access with locking and transactions, and supports backup and recovery. Because all access goes through the DBMS, data is controlled centrally and shared safely.

Data independence

Data independence is the separation of the logical view of data used by programs from the way it is physically stored. With physical data independence, storage details such as indexes or disks can change without affecting programs. With logical data independence, new fields or tables can be added and existing applications still work. This reduces maintenance costs and allows the database to evolve as the organisation grows.

T Key terms from the mark schemes

Flat file
A database held as a single table or file with no relationships to other tables.

Data redundancy
The unnecessary storage of the same data item more than once.

Data inconsistency
A situation where copies of the same data item hold different values, so it is unclear which is correct.

Database management system (DBMS)
Software that allows a database to be defined, created, queried, updated and controlled, including security, integrity and concurrent access.

Data independence
The ability to change the way data is stored or structured without having to change the application programs that use it.

Relational database
A database in which data is stored in tables (relations) linked to each other through primary and foreign keys.

Exam focus

'Explain' questions need a point plus its consequence: 'data is stored once (1) so an update only has to be made in one place, avoiding inconsistency (1)'. Do not confuse redundancy (duplication) with inconsistency (disagreeing copies) – examiners treat them as separate points. When asked for DBMS functions, give distinct functions (security, integrity, concurrency, backup, data dictionary, query language) rather than several examples of the same one. Avoid vague words such as 'easier' or 'faster' without saying why.

? Check yourself

1. Define the term data redundancy. [1]
2. Explain how data redundancy can lead to data inconsistency. [2]
3. Describe four functions of a database management system. [4]

Answers: 1. Data redundancy is the unnecessary storing of the same data more than once (e.g. in several records or files) (1). Accept: duplication of data. 2. When the same data is held in several places, an update may be made to one copy but not the others (1) / so the copies no longer agree and the organisation cannot tell which value is correct, e.g. a member's address changed on one booking but not the rest (1). 3. Any four from, with a description: stores, retrieves and updates data, usually via a query language such as SQL (1) / maintains the data dictionary of metadata about tables, fields, types and relationships (1) / enforces integrity: validation rules, entity and referential integrity (1) / provides security through user accounts, passwords and access rights to tables, views or fields (1) / controls concurrent access, e.g. record locking so multi-user updates do not conflict (1) / provides backup and recovery after failure (1) / provides data independence by separating programs from stored data via the schema (1).

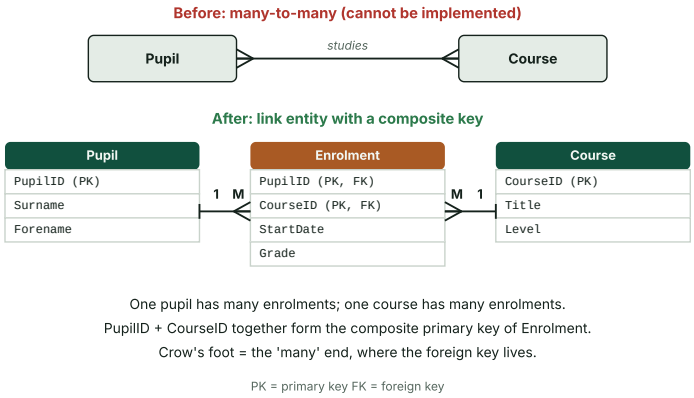
Entity-relationship modelling

IS02 FACT FILE

Specification: students should be able to

- Describe entities, attributes and relationships
- Explain primary, foreign, composite and candidate keys
- Draw and interpret entity-relationship (ER) diagrams showing 1:1, 1:M and M:N relationships
- Resolve many-to-many relationships using a link entity

Key words: entity, attribute, relationship, primary key, foreign key, composite key, candidate key, one-to-many, many-to-many, link entity, crow's foot, degree



A many-to-many relationship between Pupil and Course resolved by an Enrolment link entity with a composite key, giving two one-to-many relationships

Entities, attributes and keys

An entity is any object, person, place or event about which data is held; its attributes are the data items describing it. Every entity needs a primary key that uniquely identifies each occurrence. A foreign key is an attribute in one table that is the primary key of another, creating the link between them. Where no single attribute is unique, a composite key made from two or more attributes is used. Any attribute that could be chosen as the primary key is a candidate key.

Relationships

The degree of a relationship describes how many occurrences of one entity relate to another: one-to-one, one-to-many or many-to-many. In crow's foot notation the many end is drawn with three prongs. In a one-to-many relationship the foreign key goes in the table at the many end. A one-to-one relationship can often be merged into a single table unless the data is optional or needs separate security.

Resolving many-to-many

A many-to-many relationship cannot be implemented directly because a field cannot hold a list of values. It is resolved by introducing a link entity whose primary key is usually a composite of the two original primary keys. The link entity turns one M:N relationship into two 1:M relationships and is often where useful attributes live, such as the date of an enrolment or the grade achieved.

T Key terms from the mark schemes

Entity
An object, person, place or event about which data is stored; it becomes a table in a relational database.

Attribute
A property or characteristic of an entity; it becomes a field in a table.

Primary key
An attribute, or combination of attributes, that uniquely identifies each record in a table.

Foreign key
An attribute in one table that is the primary key of another table and is used to link the two tables.

Composite key
A primary key made up of two or more attributes which together uniquely identify a record.

Link entity
An entity introduced to resolve a many-to-many relationship into two one-to-many relationships.

Exam focus

You may be asked to draw an ER diagram from a scenario or complete one. Marks are given for each correct entity, each correct relationship with the crow's foot at the correct end, and for resolving M:N with a correctly named link entity. When asked to 'explain why' a link entity is needed, state that an M:N relationship cannot be implemented in a relational database and that the link entity creates two 1:M relationships. Underline primary keys and use an asterisk or the words 'foreign key' – be consistent.

? Check yourself

1. State what is meant by a composite key. [1]
2. Explain the purpose of a foreign key. [2]
3. A library allows each member to borrow many books, and over time each book is borrowed by many members. Draw an ER diagram showing how this relationship would be implemented in a relational database. [4]

Answers: 1. A composite key is a primary key made up of two or more attributes which together uniquely identify each record, e.g. (PupilID, CourseID) in Enrolment (1). 2. A foreign key is an attribute in one table that is the primary key of another table (1) / it is used to create the link (relationship) between the two tables, so related records can be joined, and allows referential integrity to be enforced (1). 3. Three entities drawn: Member, Book and a link entity, e.g. Loan (1) / Member to Loan is one-to-many with the crow's foot at Loan (1) / Book to Loan is one-to-many with the crow's foot at Loan (1) / no direct many-to-many line remains between Member and Book, i.e. M:N correctly resolved (1). Diagram: Member 1 ——< Loan >—— 1 Book. Accept any sensible name for the link entity (Borrowing, Issue).

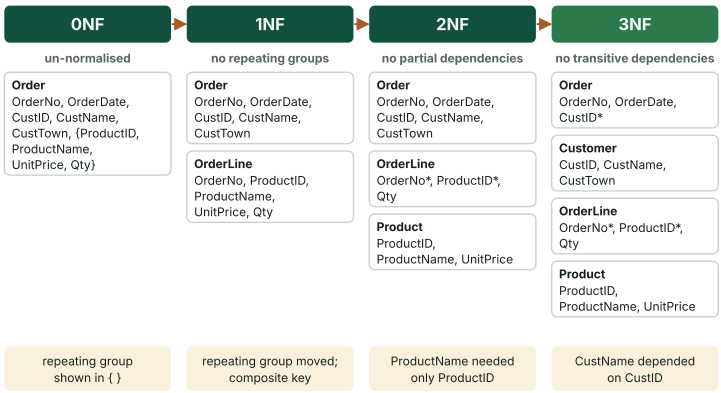
Normalisation to third normal form

IS03 FACT FILE

Specification: students should be able to

- Explain the purpose of normalisation
- Normalise data from un-normalised form (0NF) to first, second and third normal form
- Identify partial and transitive dependencies

Key words: normalisation, 0NF, first normal form, second normal form, third normal form, repeating group, atomic, functional dependency, partial dependency, transitive dependency, update anomaly, insertion anomaly



Fermanagh Fresh order data normalised from 0NF through 1NF and 2NF to 3NF, showing which dependency is removed at each stage

Purpose of normalisation

Normalisation is a step-by-step process for structuring data into well-designed relations. It minimises redundancy and so prevents update, insertion and deletion anomalies. An update anomaly occurs when the same fact stored in several rows is changed in some but not all of them. An insertion anomaly is being unable to add data because other unrelated data is missing. A deletion anomaly is losing useful data when another record is removed.

The three normal forms

First normal form removes repeating groups so each field holds a single atomic value; the repeating group goes into a new table carrying the original key, which becomes part of a composite key. Second normal form removes partial dependencies, where a non-key attribute depends on only part of a composite key. Third normal form removes transitive dependencies, where a non-key attribute depends on another non-key attribute. A relation with a single-attribute key in 1NF is automatically in 2NF.

Key terms from the mark schemes

Normalisation
A formal process of organising data into relations to minimise redundancy and prevent update, insertion and deletion anomalies.

First normal form (1NF)
A relation with no repeating groups in which every attribute holds a single atomic value.

Second normal form (2NF)
A relation in 1NF with no partial dependencies on a composite key.

Third normal form (3NF)
A relation in 2NF with no transitive dependencies, so every non-key attribute depends only on the primary key.

Partial dependency
Where a non-key attribute depends on only part of a composite primary key.

Transitive dependency
Where a non-key attribute depends on another non-key attribute rather than directly on the primary key.

Update anomaly
An inconsistency caused when a fact stored in several places is changed in some but not all of them.

Exam focus

Fermanagh Fresh order forms in 0NF become Customer(CustID, CustName, CustTown), Order(OrderNo, OrderDate, CustID*), Product(ProductID, ProductName, UnitPrice) and OrderLine(OrderNo*, ProductID*, Qty). ProductName was partially dependent on ProductID alone; CustName was transitively dependent on CustID. Each fact is now held once, and the four tables can be rejoined using SQL whenever a full order is needed.

? Check yourself

1. State two reasons why data is normalised. [2]
2. Define a partial dependency. [2]
3. Define a transitive dependency and give an example. [2]

Answers: 1. Any two from: to minimise / reduce data redundancy (1) / to prevent update anomalies and so data inconsistency (1) / to prevent insertion anomalies (1) / to prevent deletion anomalies (1) / to make the database easier to maintain and change (1). 2. A partial dependency exists where a non-key attribute depends on only part of a composite primary key (1) / rather than on the whole key, e.g. in OrderLine(OrderNo, ProductID, ProductName, Qty) ProductName depends only on ProductID (1). 3. A transitive dependency is where a non-key attribute depends on another non-key attribute rather than directly on the primary key (1) / e.g. in Order(OrderNo, OrderDate, CustID, CustName), CustName depends on CustID, which is not the key (1). Accept any correct example, e.g. TutorName depends on TutorID.

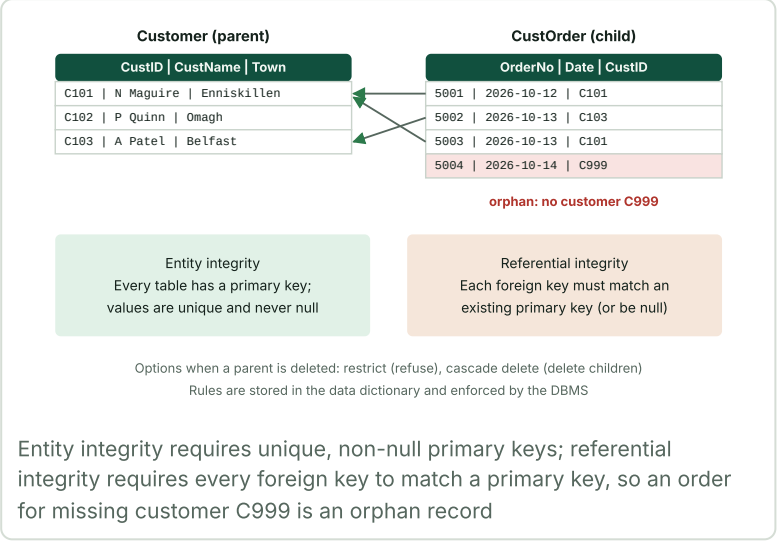
Data integrity: data dictionary, integrity rules, validation and indexing

IS04 FACT FILE

Specification: students should be able to

- Describe the purpose and contents of a data dictionary
- Explain entity integrity and referential integrity
- Describe how validation maintains data integrity
- Explain the purpose of indexing and its trade-offs

Key words: data integrity, data dictionary, metadata, entity integrity, referential integrity, cascade delete, orphan record, validation, verification, index, secondary key



Entity integrity requires unique, non-null primary keys; referential integrity requires every foreign key to match a primary key, so an order for missing customer C999 is an orphan record

Data dictionary

A data dictionary is a central store of metadata – data about the data. For every table it records the field names, data types and sizes, primary and foreign keys, validation rules, default values, indexes and relationships, and often the users who have access. Developers use it to understand the database and keep naming consistent; the DBMS uses it to enforce rules. Because the rules are held once, every application is forced to obey them.

Entity and referential integrity

Entity integrity requires every table to have a primary key whose values are unique and never null, so every record can be identified. Referential integrity requires every foreign key to match an existing primary key in the related table, preventing orphan records. The DBMS can enforce it by restricting deletions, by cascading deletions to child records or by cascading updates of key values. The right choice depends on the business rules.

Indexing

An index is a sorted data structure that maps values of one or more fields to the location of the records. It allows the DBMS to find records, sort and join tables much faster than by a full table scan, especially in large tables. However, indexes take extra storage and must be updated whenever a record is inserted, changed or deleted, so too many indexes slow down data entry. Good practice is to index fields that are frequently searched or used in joins.

T Key terms from the mark schemes

Data dictionary
A central store of metadata describing every table, field, data type, key, validation rule, index and relationship in a database.

Metadata
Data that describes other data, such as a field's name, type and size.

Entity integrity
The rule that every table has a primary key whose values are unique and never null.

Referential integrity
The rule that every foreign key value must match an existing primary key value in the related table, preventing orphan records.

Index
A sorted data structure of key values and pointers that lets the DBMS locate records quickly without scanning the whole table.

Cascade delete
A referential integrity option in which deleting a parent record automatically deletes its related child records.

Exam focus

Distinguish validation (checks data is reasonable and obeys rules) from verification (checks data matches the original source); neither guarantees accuracy. For integrity questions, name the rule and explain its effect in context: 'referential integrity would stop an Order being added for a CustID that does not exist in Customer'. For 'evaluate the use of indexes' give the benefit (faster search/join) and the cost (storage, slower updates) and reach a conclusion.

? Check yourself

1. State three items of metadata held in a data dictionary. [3]
2. Explain what is meant by entity integrity. [2]
3. Explain how referential integrity would prevent an error in a database containing Customer and Order tables. [3]

Answers: 1. Any three from: table names / field names (1) / data types (1) / field sizes / lengths (1) / primary keys (1) / foreign keys and relationships (1) / validation rules (1) / default values (1) / indexes (1) / users and access rights (1) / descriptions / formats (1). 2. Every table must have a primary key, and each primary key value must be unique (1) / and the primary key must never be null (empty), so every record can be identified unambiguously (1). 3. Referential integrity means every foreign key value must match an existing primary key value in the related table (1) / so the DBMS would reject an Order whose CustID does not exist in the Customer table, e.g. an order for C999 (1) / preventing orphan records; it would also stop (restrict) the deletion of a customer who still has orders, or cascade the change (1).

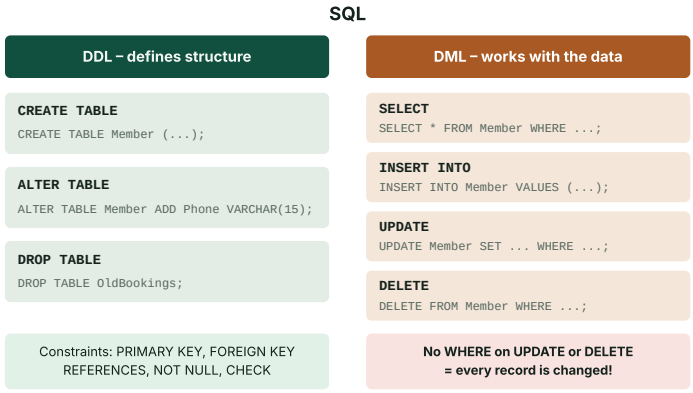
SQL data definition and data manipulation

IS05 FACT FILE

Specification: students should be able to

- Use SQL data definition language (DDL) to create tables with data types, primary keys and foreign keys
- Use SQL data manipulation language (DML) to insert, update and delete records

Key words: SQL, DDL, DML, CREATE TABLE, data type, PRIMARY KEY, FOREIGN KEY, NOT NULL, INSERT INTO, UPDATE, DELETE, constraint



SQL split into data definition language for structure and data manipulation language for data, with an example of each command

DDL and DML

SQL has a data definition language for creating and changing the structure of a database and a data manipulation language for working with the data itself. CREATE TABLE defines a table's fields, data types and constraints; ALTER TABLE changes an existing structure and DROP TABLE removes it. INSERT adds records, UPDATE changes values in existing records, DELETE removes records and SELECT retrieves them. Being declarative, SQL lets the DBMS choose the most efficient way to carry out each command.

Constraints

Constraints written in CREATE TABLE let the DBMS enforce integrity automatically. PRIMARY KEY enforces uniqueness and prevents nulls in the key. FOREIGN KEY with REFERENCES ensures each foreign key matches a primary key in the parent table. NOT NULL acts as a presence check and CHECK enforces a range or other condition. Because constraints live in the schema, every application that uses the database must obey them.

UPDATE and DELETE safely

UPDATE and DELETE apply to every record that satisfies the WHERE clause, so the condition must be precise, normally using the primary key. A missing WHERE affects the whole table. Good practice is to run a SELECT with the same WHERE first to see which rows will be affected, and to use a transaction so the change can be rolled back if it is wrong. Users should only be granted the permissions they need.

T Key terms from the mark schemes

SQL
Structured Query Language: the standard declarative language for defining, querying and manipulating relational databases.

Data definition language (DDL)
The part of SQL used to create and change database structure, e.g. CREATE TABLE, ALTER TABLE, DROP TABLE.

Data manipulation language (DML)
The part of SQL used to retrieve and change data, e.g. SELECT, INSERT, UPDATE, DELETE.

Constraint
A rule defined on a table, such as PRIMARY KEY, FOREIGN KEY, NOT NULL or CHECK, that the DBMS enforces on every change.

Exam focus

Write SQL in the order the syntax expects and use the exact table and field names from the question. Marks are typically given per clause: for example one mark for UPDATE table, one for the SET clause and one for the correct WHERE. Put text values in single quotes and do not quote numbers. In CREATE TABLE questions, choose sensible types and sizes and include PRIMARY KEY and FOREIGN KEY … REFERENCES … where asked. Minor syntax slips are often tolerated if the logic is clear, but missing WHERE clauses are not.

? Check yourself

1. State the difference between DDL and DML, giving one SQL command for each. [2]
2. Write an SQL statement to create a table Member with fields MemberID (6 characters, primary key), Surname (up to 30 characters, required) and JoinDate (date). [4]
3. Write an SQL statement to add a member with ID M0042, surname Corrigan, who joined on 1 September 2026. [2]

Answers: 1. DDL (data definition language) defines / changes the structure of the database, e.g. CREATE TABLE, ALTER TABLE or DROP TABLE (1) / DML (data manipulation language) works with the data itself, e.g. SELECT, INSERT, UPDATE or DELETE (1). 2. Model answer:

```
CREATE TABLE Member (
  MemberID CHAR(6) NOT NULL,
  Surname  VARCHAR(30) NOT NULL,
  JoinDate DATE,
  PRIMARY KEY (MemberID)
);
```

Marks: CREATE TABLE Member with fields in brackets, correct syntax (1) / MemberID CHAR(6) (accept VARCHAR(6)) (1) / Surname VARCHAR(30) NOT NULL (1) / JoinDate DATE and PRIMARY KEY (MemberID) (1). 3. Model answer:

```
INSERT INTO Member (MemberID, Surname, JoinDate)
VALUES ('M0042', 'Corrigan', '2026-09-01');
```

Marks: INSERT INTO Member with field list (accept omitted field list if values are in table order) (1) / VALUES with text and date in quotes in matching order (1). Accept #01/09/2026# (Access) or '01/09/2026'.

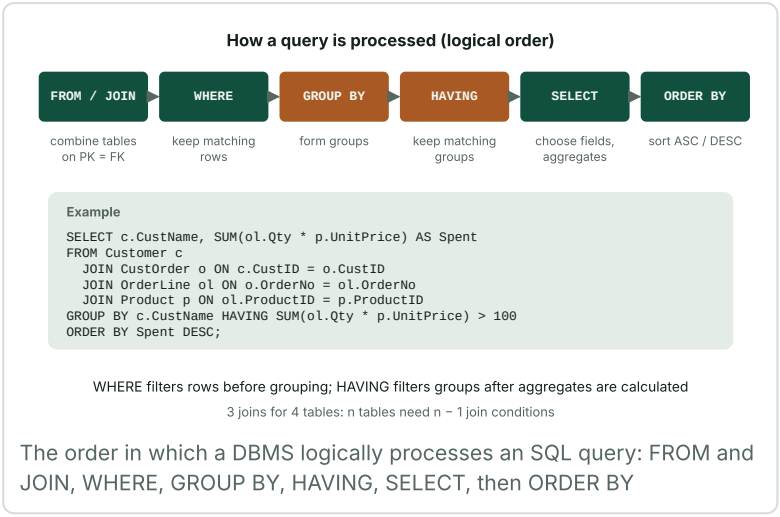
SQL queries: SELECT, aggregates, grouping and joins

IS06 FACT FILE

Specification: students should be able to

- Write SQL SELECT queries using WHERE, ORDER BY, AND, OR, LIKE and BETWEEN
- Use aggregate functions with GROUP BY
- Write queries that join two or three tables

Key words: SELECT, FROM, WHERE, ORDER BY, LIKE, wildcard, BETWEEN, aggregate function, GROUP BY, HAVING, INNER JOIN, alias



The order in which a DBMS logically processes an SQL query: FROM and JOIN, WHERE, GROUP BY, HAVING, SELECT, then ORDER BY

Single-table queries

A SELECT query names the fields to display, the table in FROM, an optional WHERE condition and an optional ORDER BY. Conditions use comparison operators and combine with AND, OR and NOT, with brackets controlling the order. LIKE matches patterns using % for any sequence of characters and _ for a single character. BETWEEN is inclusive of both limits, and IN checks whether a value is in a list.

Aggregates and grouping

Aggregate functions calculate one value from many rows: COUNT, SUM, AVG, MIN and MAX. Used alone they summarise the whole result; with GROUP BY they produce one row per group. Any field shown that is not inside an aggregate must be listed in GROUP BY. WHERE filters individual rows before grouping, while HAVING filters the groups after the aggregates have been calculated. AS gives a calculated column a readable name.

Joins

Because normalised data is split across tables, queries often need joins. An inner join returns only rows where the foreign key in one table matches the primary key in the other. Joining three tables needs two join conditions, following the path of relationships in the ER diagram, for example Customer to CustOrder to OrderLine. If a join condition is omitted, the result is a Cartesian product in which every row of one table is paired with every row of the other.

T Key terms from the mark schemes

Aggregate function
An SQL function that calculates a single value from a set of rows, such as COUNT, SUM, AVG, MIN or MAX.

GROUP BY
An SQL clause that groups rows with the same value in a field so aggregate functions are calculated for each group.

HAVING
An SQL clause that filters groups after aggregation, in the way WHERE filters individual rows.

Join
An operation that combines rows from two or more tables by matching a foreign key with the related primary key.

Wildcard
A symbol used with LIKE to match unknown characters: % for any sequence and _ for exactly one character.

Exam focus

Read the question carefully for exactly which fields to display and in what order to sort. Marks are usually awarded per clause: SELECT fields (1), FROM tables (1), join condition (1), each criterion (1), ORDER BY (1). Use the table and field names given, quote text, and write dates in the format shown. When asked to 'state the output' of a query on a given table, list the rows in the order the query would return them, with only the selected fields.

? Check yourself

1. Write an SQL query to list the Surname and Forename of all pupils in YearGroup 14, in alphabetical order of Surname. [3]
2. Write an SQL query to display all products whose name begins with 'Organic' and whose price is between £1 and £3. [3]
3. Explain the difference between WHERE and HAVING. [2]

Answers: 1. Model answer:
SELECT Surname, Forename
FROM Pupil
WHERE YearGroup = 14
ORDER BY Surname;
Marks: SELECT Surname, Forename FROM Pupil (1) / WHERE YearGroup = 14 (1) / ORDER BY Surname (ASC) (1). Accept any sensible table name. 2. Model answer:
SELECT *
FROM Product
WHERE ProductName LIKE 'Organic%'
 AND UnitPrice BETWEEN 1 AND 3;
Marks: SELECT * FROM Product (1) / ProductName LIKE 'Organic%' (accept * wildcard in Access) (1) / AND UnitPrice BETWEEN 1 AND 3 (accept >= 1 AND <= 3) (1). 3. WHERE filters individual rows before they are grouped / aggregated (1) / HAVING filters groups after GROUP BY and aggregation, so it can use aggregate functions such as COUNT(*) (1).

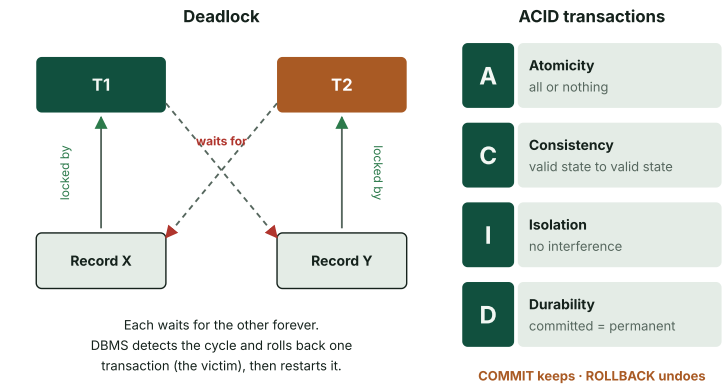
Multi-user databases: transactions, ACID, locking and distributed databases

IS07 FACT FILE

Specification: students should be able to

- Explain transactions and the ACID properties
- Describe record locking, deadlock and how deadlock is handled
- Explain commit and rollback
- Compare client–server and distributed databases

Key words: transaction, ACID, atomicity, consistency, isolation, durability, commit, rollback, record locking, deadlock, lost update, client–server, distributed database



Deadlock: transaction T1 holds a lock on record X and waits for Y while T2 holds Y and waits for X; the DBMS rolls one back. ACID properties listed alongside

Transactions and ACID

A transaction is a sequence of operations treated as a single unit. The ACID properties make transactions reliable. Atomicity means all or none of the operations take effect. Consistency means a transaction takes the database from one valid state to another, obeying all integrity rules. Isolation means concurrent transactions do not interfere and give the same result as running one after another. Durability means committed changes are permanent, even after a crash, usually because they are written to a transaction log.

Concurrency control

When several users update the same data at once, the lost update problem can occur: one user's change overwrites another's. Record locking prevents this by giving a transaction exclusive access to a record until it commits or rolls back. Locking can lead to deadlock, where two transactions each hold a lock the other needs. A DBMS detects deadlock and rolls back one transaction, or prevents it by ordering locks or requesting all locks at the start.

Client–server and distributed

In a client–server database the data and DBMS are held centrally and clients send queries, receiving only the results; this gives central control and lower network traffic but creates a single point of failure. A distributed database spreads one logical database across several sites using fragmentation and replication. It gives fast local access and resilience, but keeping replicas consistent, securing several sites and committing transactions across sites is complex.

T Key terms from the mark schemes

Transaction
A sequence of database operations treated as a single logical unit that must either complete entirely or not at all.

ACID
Atomicity, consistency, isolation and durability: the properties that guarantee database transactions are processed reliably.

Record locking
A concurrency control technique that prevents other users updating a record while one transaction is using it.

Deadlock
A situation where two or more transactions each hold a lock the other needs, so none can proceed.

Rollback
Undoing all changes made by a transaction since it began, returning the database to its previous consistent state.

Distributed database
A single logical database whose data is stored at several sites linked by a network but appears to users as one database.

Exam focus

Each ACID property needs a definition plus an applied example to gain full marks; use the scenario given. For locking questions describe the sequence of events with times or users (A reads, B reads, A writes, B writes) – timelines earn marks. Deadlock answers need the cause (each holds a lock the other wants) and the resolution (DBMS detects, rolls back one). For 'discuss' questions on distributed databases balance local speed and resilience against complexity, security and consistency, and conclude.

? Check yourself

1. Define the term transaction in the context of databases. [2]
2. Explain atomicity and durability, using a bank transfer as an example. [4]
3. Describe how the lost update problem can occur and how record locking prevents it. [4]

Answers: 1. A transaction is a group of one or more database operations (e.g. updates) (1) / that must be treated as a single logical unit of work: either all succeed (COMMIT) or none are kept (ROLLBACK) (1). 2. Atomicity: all operations in the transaction happen or none do (1) / e.g. the £50 is never debited from savings without also being credited to the current account; if the credit fails the debit is rolled back (1) / Durability: once the transaction is committed its changes are permanent (1) / e.g. the transfer is still recorded after a power cut or server crash/restart, because it was written to the log/disk (1). 3. Two users / transactions read the same record at the same time, e.g. both read Balance = 100 (1) / each changes their copy and writes it back, e.g. A writes 70 and then B writes 80, so A's update is overwritten (lost); the balance should be 50 (1) / With record locking, the first transaction to read the record for update locks it (exclusive lock) (1) / the second must wait until the lock is released at COMMIT/ROLLBACK, so it reads the updated value (70) and writes 50 (1).

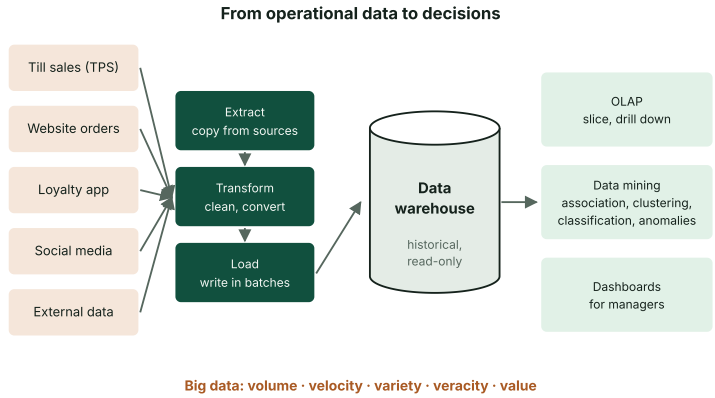
Big data, data warehousing, data mining and analytics

IS08 FACT FILE

Specification: students should be able to

- Describe the characteristics of big data
- Explain data warehousing, including extract–transform–load
- Describe data mining techniques and the use of analytics in organisations
- Discuss privacy issues arising from big data and data mining

Key words: big data, volume, velocity, variety, veracity, data warehouse, ETL, OLAP, data mining, association, clustering, predictive analytics



Data from operational sources is extracted, transformed and loaded into a data warehouse, then analysed with OLAP and data mining to support decisions

Big data

Big data describes data sets whose volume, velocity and variety make them too large or complex for traditional database tools. Veracity (accuracy and trustworthiness) and value (the useful insight that can be extracted) are often added. Much big data is unstructured, such as text, images and sensor streams, so it is stored and processed across clusters of servers in parallel, often in NoSQL databases rather than relational tables.

Data warehouses

A data warehouse is a central repository of integrated, historical data drawn from many operational systems. Data is extracted, transformed into a consistent and clean format, and loaded (ETL). Warehouses are subject-oriented, time-variant and non-volatile, and are usually denormalised to make complex analytical queries faster. Managers use OLAP tools to slice data and drill down, for example from yearly sales to monthly sales by branch.

Data mining and analytics

Data mining uses statistical and machine learning techniques to discover hidden patterns in large data sets. Common techniques are association rules, classification, clustering, anomaly detection and regression. Results support descriptive, predictive and prescriptive analytics, helping organisations target marketing, manage stock, detect fraud and plan services. The benefits must be weighed against cost, data quality and the privacy of the individuals whose data is mined.

T Key terms from the mark schemes

Big data
Data sets so large, fast-moving or varied that they cannot be processed efficiently with traditional database tools.

ETL
Extract, transform, load: the process of copying data from source systems, cleaning and converting it, and loading it into a data warehouse.

OLAP
Online analytical processing: tools that let users analyse multidimensional data, for example by slicing and drilling down.

Data warehouse
A large store of integrated, historical, read-only data from many sources, organised for analysis and decision-making.

Data mining
The process of analysing large data sets to discover previously unknown patterns, relationships and trends.

Exam focus

Extended questions on big data and data mining are often marked with levels of response including quality of written communication. Define the terms precisely, apply them to the scenario, and balance benefits against issues such as privacy, consent, security and bias, linking to the Data Protection Act 2018 and UK GDPR where relevant. Avoid listing techniques without saying what each finds. A clear conclusion that weighs the points is needed for the top band.

? Check yourself

1. Describe three characteristics of big data. [3]
2. Explain the stages of ETL used when loading a data warehouse. [3]
3. Give two differences between an operational database and a data warehouse. [2]

Answers: 1. Any three from, each described: Volume – huge quantities of data, terabytes to petabytes (1) / Velocity – data arrives very quickly and must be processed fast, often in real time (1) / Variety – many types and formats, structured and unstructured such as text, images and video (1) / Veracity – data may be of uncertain accuracy or trustworthiness (1) / Value – useful insight can be gained from it (1). 2. Extract: data is copied from the various operational source systems (1) / Transform: data is cleaned and made consistent, e.g. converting formats, removing duplicates and errors (1) / Load: the transformed data is loaded into the data warehouse for analysis (1). 3. Any two from: operational databases support daily transactions, a warehouse supports analysis and decision-making (1) / operational data is current and changing, warehouse data is historical and read-only (non-volatile) (1) / operational databases are normalised to 3NF, warehouses are often denormalised (e.g. star schema) (1) / operational queries are many short updates, warehouse queries are few, large, complex reads (1) / operational users are staff, warehouse users are managers/analysts (1) / warehouse integrates data from many sources (1).

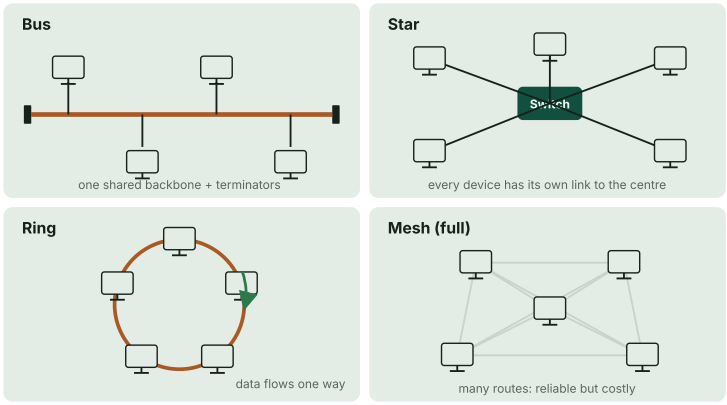
Network types, topologies and hardware

IS09 FACT FILE

Specification: students should be able to

- Describe LAN, WAN, PAN and MAN networks
- Compare bus, star, ring and mesh topologies
- Describe the function of network hardware: NIC, switch, router, wireless access point and gateway

Key words: LAN, WAN, PAN, MAN, topology, bus, star, ring, mesh, NIC, switch, router, wireless access point, gateway



The four network topologies: bus with a shared backbone, star with a central switch, ring as a closed loop, and mesh with multiple links

Network types

Networks are classified by the area they cover. A personal area network links a person's own devices over a few metres, often by Bluetooth. A local area network covers one site and is owned and managed by the organisation. A metropolitan area network covers a town or city, linking sites such as council buildings or university campuses. A wide area network spans countries, using telecommunications links leased from third parties; the internet is the largest WAN.

Topologies

A topology is the arrangement of devices and links. A bus shares one backbone cable, so it is cheap but suffers collisions and fails if the backbone breaks. A ring passes data around a loop in one direction with no collisions, but one break can stop it. A star connects each device to a central switch, isolating cable faults and giving high performance, though the switch is a single point of failure. A mesh gives multiple routes, providing resilience at high cost.

Hardware

A network interface card provides the physical connection and a unique MAC address. A switch forwards frames within a LAN to the correct port using MAC addresses. A router joins different networks and forwards packets between them using IP addresses and routing tables. A wireless access point connects wireless devices to a wired network. A gateway joins networks that use different protocols, translating between them.

T Key terms from the mark schemes

Local area network (LAN)
A network covering a single site or building, with infrastructure owned and managed by the organisation.

Wide area network (WAN)
A network covering a large geographical area, connecting LANs using third-party telecommunications links.

Topology
The physical or logical arrangement of the devices and links in a network.

Switch
A device that connects devices within a LAN and forwards each frame only to the port of its destination, using MAC addresses.

Router
A device that connects different networks and forwards packets towards their destination using IP addresses and routing tables.

Gateway
A device that connects two networks using different protocols and translates between them.

Exam focus

Many questions ask you to recommend and justify a topology or device for a scenario. A justification needs a feature and its benefit in context, for example 'a star topology means a cable fault in one guest room only affects that room, so the rest of the hotel stays online'. When comparing, use the same criteria for each option (reliability, performance, cost, scalability). Do not describe a switch as broadcasting data – that is a hub – and remember routers use IP addresses while switches use MAC addresses.

? Check yourself

1. State one difference between a LAN and a WAN. [1]
2. Describe the function of a switch in a star network. [2]
3. Explain the difference between a router and a gateway. [2]

Answers: 1. Any one from: a LAN covers a small area such as one site or building, while a WAN covers a large geographical area such as countries (1) / a LAN's infrastructure is owned by the organisation, while a WAN uses third-party telecoms links / leased lines (1). 2. The switch connects all the devices in the star; each device has its own cable to it (1) / it learns the MAC address of each device and forwards each frame only to the port of the destination device, rather than to all devices, reducing traffic and collisions (1). 3. A router connects different networks (e.g. a LAN to the internet) and forwards packets using IP addresses and routing tables to choose the best route (1) / a gateway connects networks that use different protocols, translating between them (1).

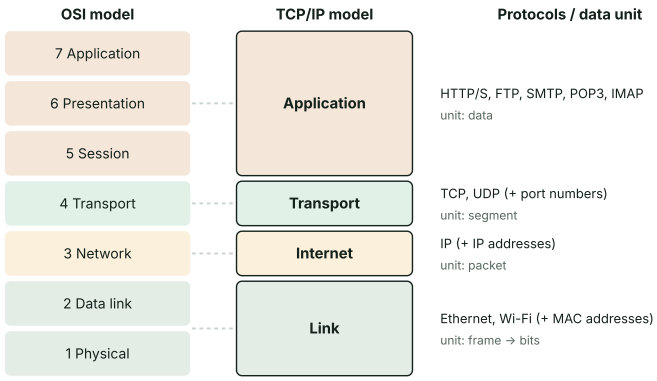
Layered network models and protocols

IS10 FACT FILE

Specification: students should be able to

- Explain why layered models are used in networking
- Describe the layers of the OSI and TCP/IP models
- Describe the purpose of common protocols: HTTP/HTTPS, FTP, SMTP, POP3, IMAP, TCP, UDP, IP, Ethernet and Wi-Fi

Key words: protocol, layer, OSI model, TCP/IP model, encapsulation, HTTP, HTTPS, FTP, SMTP, IMAP, TCP, UDP, IP, Ethernet



The seven OSI layers mapped to the four TCP/IP layers, with example protocols and the header added at each layer during encapsulation

Why use layers

Networking is divided into layers so that each deals with one part of the problem. Each layer offers services to the one above and relies on the one below, communicating through standard interfaces. This means one layer can be replaced, such as wired Ethernet with Wi-Fi, without changing the others. Layers make networks easier to design, develop, teach and troubleshoot, and open standards let hardware and software from different manufacturers work together.

OSI and TCP/IP

The seven-layer OSI model is a reference model: physical, data link, network, transport, session, presentation and application. The four-layer TCP/IP model is the practical model used on the internet: link, internet, transport and application, with the top three OSI layers combined into the application layer. Data is encapsulated on the way down, each layer adding a header, and de-encapsulated on the way up at the receiver.

Key protocols

HTTP transfers web pages and HTTPS does the same over an encrypted TLS connection. FTP transfers files. SMTP sends email, while POP3 downloads it to one device and IMAP keeps it on the server synchronised across devices. TCP provides reliable, ordered, connection-oriented delivery; UDP provides fast, connectionless delivery without guarantees. IP addresses and routes packets. Ethernet and Wi-Fi handle transmission across wired and wireless links.

T Key terms from the mark schemes

Protocol
A set of rules that governs how data is formatted, transmitted and received so devices can communicate.

OSI model
A seven-layer reference model of network communication: physical, data link, network, transport, session, presentation and application.

TCP/IP model
The four-layer model used by the internet: link, internet, transport and application.

Encapsulation (networking)
The process by which each layer adds its own header to data as it passes down the layers before transmission.

TCP
Transmission Control Protocol: a connection-oriented transport protocol giving reliable, ordered delivery using acknowledgements and retransmission.

UDP
User Datagram Protocol: a connectionless transport protocol that is fast but does not guarantee delivery or order.

Exam focus

When asked for an advantage of layering, explain the consequence: 'a layer can be changed without affecting other layers, so a new transmission medium can be introduced without rewriting applications'. When describing a protocol, state its purpose precisely rather than just expanding the acronym. For TCP versus UDP choices, justify using the application: lost data in a file download is unacceptable (TCP) but a late video frame is useless anyway (UDP). Learn which layer each protocol belongs to.

? Check yourself

1. Define the term protocol. [1]
2. Explain two advantages of using a layered model for network communication. [4]
3. Name the four layers of the TCP/IP model in order from the top. [2]

Answers: 1. A protocol is a set of rules that governs how data is formatted, transmitted and received between devices (1). 2. Any two advantages, each stated (1) and explained (1): one layer can be changed or replaced without affecting the others (1) / e.g. Wi-Fi can replace Ethernet at the link layer without changing TCP or HTTP (1) / complex communication is split into smaller, manageable tasks (1) / so each layer is easier to design, develop, teach and troubleshoot (1) / layers follow open standards (1) / so equipment and software from different manufacturers can interoperate (1). 3. Application, Transport (1) / Internet (accept Network), Link (accept Network access / Network interface) (1). All four in the correct order from the top for both marks; award 1 mark if all four are named but the order is wrong.

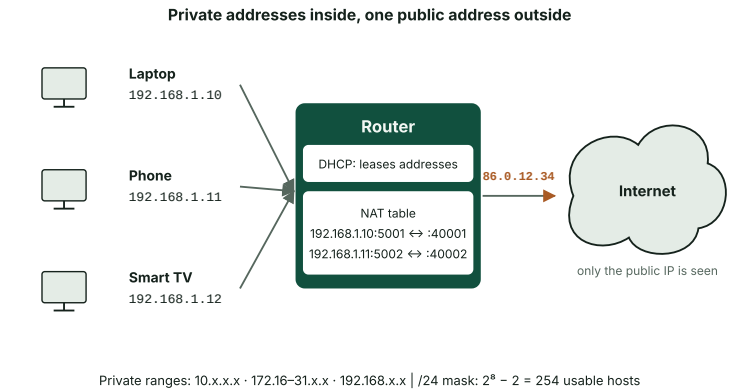
IP addressing, switching and routing

IS11 FACT FILE

Specification: students should be able to

- Describe IPv4 and IPv6 addressing, public and private addresses and the purpose of subnetting
- Explain the roles of DHCP and NAT
- Compare packet switching and circuit switching
- Describe how routing works

Key words: IP address, IPv4, IPv6, private address, public address, subnet mask, DHCP, NAT, packet switching, circuit switching, routing table, hop



A home network where DHCP gives devices private IP addresses and the router uses NAT to share one public IP address with the internet

IP addressing

An IP address is a logical address identifying a device on a network. IPv4 uses 32 bits written as four denary octets, giving about 4.3 billion addresses, which proved insufficient. IPv6 uses 128 bits written in hexadecimal, providing a vast address space. Private address ranges such as 192.168.0.0/16 are used inside LANs and not routed on the internet, while public addresses are unique worldwide. A subnet mask separates the network part of an address from the host part.

DHCP, NAT and subnets

DHCP automatically allocates an IP address, subnet mask, default gateway and DNS server to each device for a lease period, avoiding manual errors and duplicate addresses. NAT allows many devices with private addresses to share one public IPv4 address by rewriting addresses at the router and tracking each connection. Subnetting divides a network into smaller logical networks, reducing broadcast traffic, improving security and simplifying management.

Switching and routing

In circuit switching a dedicated path is reserved for the whole communication, giving a constant rate but wasting capacity during silence. In packet switching data is broken into packets with headers containing addresses and sequence numbers; packets travel independently, possibly by different routes, and are reassembled at the destination. Routers use routing tables to forward each packet to the next hop, so traffic can avoid congested or failed links and capacity is shared efficiently.

Key terms from the mark schemes

IP address
A logical numerical address that identifies a device on a network so data can be routed to it.

DHCP
Dynamic Host Configuration Protocol: a service that automatically assigns IP addresses and network settings to devices.

Packet switching
A method of transmission in which data is split into packets that travel independently across shared links and are reassembled at the destination.

Subnet mask
A value that identifies which bits of an IP address form the network part and which form the host part.

Network address translation (NAT)
A router technique that maps private IP addresses to a public IP address so many devices can share one public address.

Circuit switching
A method of transmission in which a dedicated path is set up and reserved between sender and receiver for the whole communication.

Exam focus

Calculation questions may ask for the number of addresses or usable hosts: find the host bits (32 minus the prefix length), calculate 2 to that power and subtract 2 for the network and broadcast addresses. Show your working for method marks. For comparison questions use matching criteria for both methods. When explaining packet switching, mention splitting into packets, headers with destination address and sequence number, independent routes, and reassembly – each is a creditable point.

? Check yourself

1. State the number of bits in an IPv4 address and in an IPv6 address. [2]
2. Explain why IPv6 was introduced. [2]
3. A network uses the subnet mask 255.255.255.192 (/26). Calculate the number of usable host addresses. Show your working. [3]

Answers: 1. IPv4: 32 bits (1) / IPv6: 128 bits (1). 2. IPv4 provides only about 4.3 billion (2^32) addresses, which have run out (1) / because of the huge growth in internet-connected devices such as smartphones and IoT devices; IPv6's 128-bit addresses give about 3.4 × 10^38 addresses (1). 3. Host bits = 32 − 26 = 6 (1) / 2^6 = 64 addresses (1) / subtract network and broadcast addresses: 64 − 2 = 62 usable hosts (1).

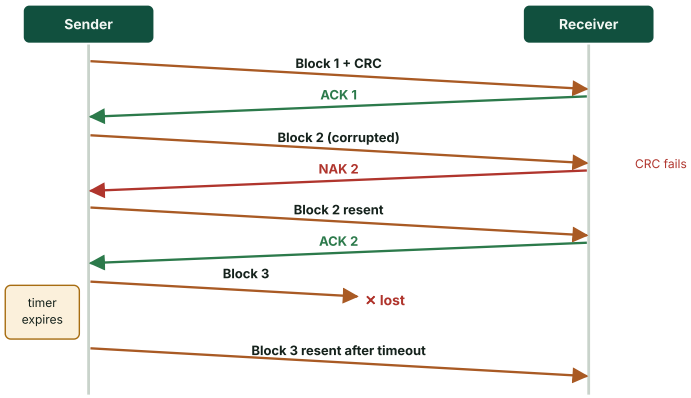
Data transmission and error control

IS12 FACT FILE

Specification: students should be able to

- Compare transmission media
- Distinguish bandwidth and bit rate and explain broadband
- Compare serial and parallel, simplex, half-duplex and full-duplex, and synchronous and asynchronous transmission
- Describe error detection (parity, checksum, CRC) and correction (ARQ)

Key words: bandwidth, bit rate, broadband, serial, parallel, simplex, duplex, synchronous, asynchronous, parity, checksum, CRC, ARQ



Automatic repeat request: block 1 is acknowledged, block 2 arrives with an error and is negatively acknowledged then resent, and block 3 is lost so a timeout triggers retransmission

Media and capacity

Twisted-pair copper is cheap and common in LANs but suffers interference and is limited to about 100 metres per run. Fibre-optic cable carries light, giving very high bandwidth over long distances, immunity to electrical interference and better security, but costs more to install. Wireless media allow mobility but are prone to interference and interception. Bandwidth is the capacity of a channel; bit rate is the number of bits actually sent each second. Transfer time equals the number of bits divided by the bit rate.

Modes of transmission

Serial transmission sends bits one after another along a single channel and works reliably over long distances. Parallel sends several bits at once but suffers skew and crosstalk, so it is used only over short distances. Simplex sends in one direction, half-duplex in both directions but not at once, and full-duplex in both directions simultaneously. Synchronous transmission uses a shared clock and sends continuous blocks; asynchronous transmission frames each character with start and stop bits.

Error detection and correction

A parity bit makes the number of 1s even or odd and detects any single-bit error but not an even number of flipped bits. A checksum is calculated from the data block and recalculated by the receiver. A cyclic redundancy check divides the data by a fixed polynomial and checks the remainder, detecting burst errors reliably. Automatic repeat request corrects errors by retransmission: the receiver sends acknowledgements, and the sender resends any block that is negatively acknowledged or whose acknowledgement does not arrive in time.

Key terms from the mark schemes

Bandwidth
The capacity of a communication channel (range of frequencies), which determines the maximum rate at which data can be transmitted.

Bit rate
The number of bits transmitted per second over a communication channel.

Full-duplex
A transmission mode in which data can travel in both directions at the same time.

Asynchronous transmission
Transmission without a shared clock, where each character is framed by start and stop bits.

Parity bit
An extra bit added to a group of bits so that the total number of 1s is even (even parity) or odd (odd parity), used to detect errors.

Automatic repeat request (ARQ)
An error-correction method in which the receiver acknowledges each block and the sender retransmits any block that is negatively acknowledged or times out.

Cyclic redundancy check (CRC)
An error-detection method in which a remainder calculated by dividing the data by an agreed polynomial is sent and checked by the receiver.

Exam focus

For calculations, write the formula, convert bytes to bits (multiply by 8), keep units consistent and give units in the answer; method marks are available even if the final answer is wrong. For error control, a full description of ARQ mentions acknowledgement, timeout and retransmission. When asked about a limitation of parity, say it cannot detect an even number of bit errors and cannot locate the error. Justify media choices in context: distance, bandwidth, interference, cost and security.

? Check yourself

1. Distinguish between bandwidth and bit rate. [2]
2. Calculate how long it would take to transfer a 250 MB file over a 20 Mbps connection. Show your working. [3]
3. Add an even parity bit to 1101001 and explain one limitation of parity checking. [3]

Answers: 1. Bandwidth is the range of frequencies a channel can carry, which determines its maximum data capacity (1) / bit rate is the number of bits actually transmitted per second (e.g. Mbps), which may be lower due to congestion, distance or interference (1). 2. Convert to bits: 250 MB × 8 = 2000 Mb (megabits) (1) / time = 2000 ÷ 20 (1) / = 100 seconds (1 minute 40 seconds) (1). Accept 104.9 s (≈105 s) if 1 MB = 1024 × 1024 bytes and 20 Mbps = 20 000 000 bps is stated. 3. 1101001 contains four 1s, which is already even (1) / so the even parity bit is 0: 11010010 (accept 01101001 if parity bit placed first) (1) / Limitation, any one: cannot detect an even number of bit errors, e.g. two bits flipped (1) / cannot identify which bit is wrong, so cannot correct the error (1).

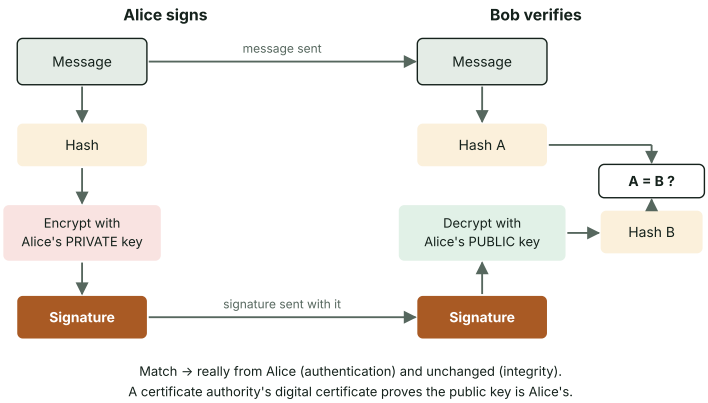
Network security: threats, firewalls, encryption and authentication

IS13 FACT FILE

Specification: students should be able to

- Describe threats to networks and data
- Explain the use of firewalls and proxy servers
- Explain symmetric and asymmetric encryption, digital signatures and digital certificates
- Describe authentication methods and the use of virtual private networks (VPNs)

Key words: malware, phishing, denial of service, firewall, proxy server, symmetric encryption, asymmetric encryption, public key, private key, digital signature, digital certificate, authentication, VPN



Creating a digital signature by hashing a message and encrypting the hash with the sender's private key, and verifying it by decrypting with the sender's public key and comparing hashes

Threats and defences

Networks face malware, phishing, hacking, denial-of-service attacks, interception of data and insider misuse. Defence in depth combines technical measures – firewalls, anti-malware, encryption, patching, authentication, VPNs – with procedures such as staff training, access rights, backups and acceptable use policies. A firewall filters traffic by rules about addresses, ports and protocols. A proxy server makes web requests on behalf of users, hiding internal addresses, filtering content and caching pages.

Encryption

Symmetric encryption uses one shared secret key for both encryption and decryption; it is fast but the key must be distributed securely. Asymmetric encryption uses a public key, which anyone can have, and a private key, kept secret by its owner; data encrypted with one can only be decrypted with the other. Because asymmetric encryption is slow, protocols such as TLS use it to exchange a symmetric session key and then encrypt the session symmetrically.

Signatures and certificates

A digital signature is created by hashing a message and encrypting the hash with the sender's private key. The receiver decrypts it with the sender's public key and compares it to a fresh hash; a match proves the sender's identity and that the message was not altered. A digital certificate issued by a certificate authority binds an organisation's identity to its public key, so users can trust that the key genuinely belongs to that organisation.

T Key terms from the mark schemes

Firewall
Hardware or software that monitors network traffic and allows or blocks it according to a set of security rules.

Proxy server
A server that acts as an intermediary between users and the internet, hiding internal addresses, filtering content and caching pages.

Symmetric encryption
Encryption in which the same secret key is used both to encrypt and to decrypt data.

Asymmetric encryption
Encryption using a mathematically related key pair: data encrypted with the public key can only be decrypted with the private key, and vice versa.

Digital signature
A hash of a message encrypted with the sender's private key, used to verify the sender's identity and that the message has not been altered.

Digital certificate
An electronic document issued by a certificate authority that binds a public key to the identity of its owner.

Virtual private network (VPN)
An encrypted connection across a public network that lets a remote user access a private network securely.

Exam focus

Be precise about keys: confidentiality uses the recipient's public key; signing uses the sender's private key. Many answers lose marks by saying 'the key' without saying whose and which. When describing a firewall, say what it checks and what it does (blocks or allows according to rules). For 'evaluate' or 'recommend' questions, match each measure to a specific threat in the scenario and mention a limitation – no single measure is sufficient.

? Check yourself

1. Describe the purpose of a firewall. [2]
2. Give two functions of a proxy server. [2]
3. Explain the key distribution problem in symmetric encryption and how asymmetric encryption overcomes it. [3]

Answers: 1. A firewall monitors incoming and outgoing network traffic (1) / and allows or blocks it according to a set of rules, e.g. based on IP addresses, ports or protocols, to prevent unauthorised access (1). 2. Any two from: makes requests to the internet on behalf of users, hiding internal IP addresses (1) / filters web content / blocks access to unsuitable sites (1) / caches popular web pages to speed up access and save bandwidth (1) / logs users' web activity (1). 3. In symmetric encryption the same secret key is used to encrypt and decrypt, so the key must somehow be shared with the receiver (1) / if it is sent over the network it could be intercepted, and anyone with it can read the messages (1) / asymmetric encryption uses a key pair: the public key can be published openly and used to encrypt, but only the matching private key, which is never shared, can decrypt (1). Accept: asymmetric encryption is used to exchange a symmetric session key securely (as in TLS).

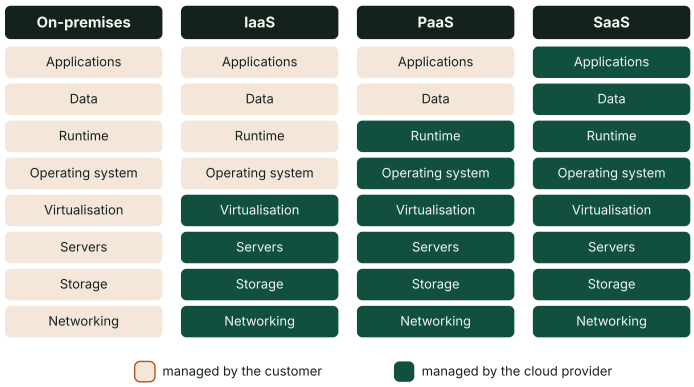
Cloud computing and virtualisation

IS14 FACT FILE

Specification: students should be able to

- Describe cloud service models: infrastructure, platform and software as a service
- Compare public, private and hybrid cloud deployment
- Evaluate the use of cloud computing by organisations
- Explain virtualisation and its benefits

Key words: cloud computing, IaaS, PaaS, SaaS, public cloud, private cloud, hybrid cloud, scalability, virtualisation, hypervisor, virtual machine, data sovereignty



Who manages each layer of the computing stack in on-premises, IaaS, PaaS and SaaS models: the customer or the cloud provider

Service models

Cloud computing provides computing resources over the internet on demand. In infrastructure as a service the provider supplies virtual servers, storage and networking, and the customer installs and manages the operating system and applications. In platform as a service the provider also manages the operating system, runtime and database, so developers only deploy their code. In software as a service the provider runs the whole application and users simply access it, usually through a browser.

Deployment models

A public cloud is owned by a provider and shared by many customers, giving low cost and high scalability. A private cloud is dedicated to one organisation, offering more control and security, which suits regulated sectors such as banking and health, but costs more. A hybrid cloud combines the two: sensitive data and core systems stay private, while public resources handle less sensitive workloads or peaks in demand.

Virtualisation

Virtualisation creates software versions of physical resources. A hypervisor divides one physical server into several isolated virtual machines, each running its own operating system. This consolidates servers, cutting hardware, power and cooling costs, and allows new servers to be created in minutes. Snapshots and live migration support backup and disaster recovery. However, a failure of the host affects all its virtual machines, and there is some performance overhead. Virtualisation underpins cloud computing.

T Key terms from the mark schemes

Cloud computing
The on-demand delivery of computing resources such as servers, storage and software over the internet, usually paid for as used.

Platform as a service (PaaS)
A cloud model in which the provider supplies the hardware, operating system and development platform so customers only build and manage their own applications.

Hybrid cloud
A cloud deployment that combines a private cloud with public cloud services, allowing data and applications to move between them.

Infrastructure as a service (IaaS)
A cloud model in which the provider supplies virtualised hardware such as servers, storage and networking, and the customer manages the operating system and software.

Software as a service (SaaS)
A cloud model in which complete application software is hosted by the provider and accessed by users over the internet.

Virtualisation
The use of software to create virtual versions of hardware, allowing several virtual machines to run on one physical computer.

Hypervisor
Software that creates, runs and manages virtual machines on a physical host computer.

Exam focus

Discussion questions on cloud computing are often extended-response with QWC. Organise your answer around benefits (scalability, cost, access, maintenance, recovery) and drawbacks (connectivity, security, legal responsibility and data location, control, lock-in, ongoing cost), always applied to the organisation in the question. Distinguish service models by who manages what. A balanced judgement such as 'a hybrid approach would suit the trust because…' is needed for the top band.

? Check yourself

1. Describe what is meant by software as a service (SaaS) and give an example. [2]
2. Explain the difference between IaaS and PaaS. [2]
3. Describe a hybrid cloud and give one reason an organisation might choose it. [2]

Answers: 1. SaaS: the complete application is provided over the internet by the provider, who manages the software, updates and infrastructure; the customer manages only their data/settings (1) / example: Microsoft 365, Gmail, Google Classroom (1). 2. IaaS provides virtualised infrastructure – servers, storage and networking – and the customer installs and manages the operating system, runtime and applications, e.g. renting virtual machines (1) / PaaS provides a platform including the OS, runtime and database, so the customer only develops and manages their own applications and data, e.g. deploying a web app to Azure App Service (1). 3. A hybrid cloud combines private cloud (or on-site) infrastructure with public cloud, with data and applications moving between them (1) / reason, any one: keep sensitive data in the private cloud while using the public cloud for less sensitive work or for peak demand (cloud bursting) (1).

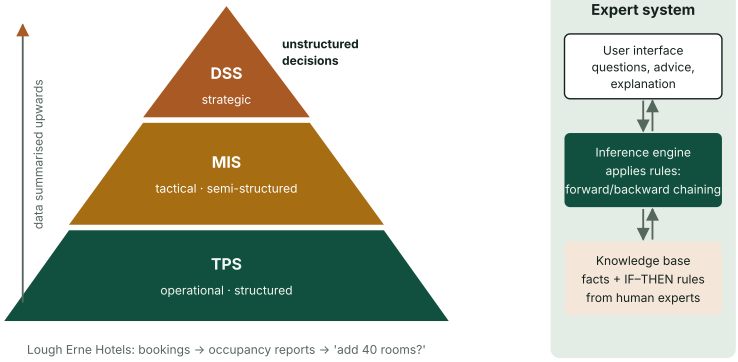
Types of information system

IS15 FACT FILE

Specification: students should be able to

- Describe the characteristics and purpose of transaction processing systems (TPS)
- Describe management information systems (MIS) and decision support systems (DSS)
- Describe expert systems, including the knowledge base, inference engine and user interface

Key words: information system, transaction processing system, batch processing, real-time processing, management information system, decision support system, what-if analysis, expert system, knowledge base, inference engine, explanation facility, strategic



The information pyramid: TPS at the operational level, MIS at the tactical level and DSS at the strategic level, with data flowing upwards; and the three components of an expert system

TPS and MIS

A transaction processing system captures and processes routine business transactions such as sales, bookings and payments, either in real time or in batches. It supports operational staff and provides the detailed data on which other systems depend. A management information system summarises TPS data into scheduled, exception and on-demand reports for middle managers, helping them monitor performance and make tactical, semi-structured decisions such as staffing levels or budgets.

Decision support systems

A decision support system helps managers tackle semi-structured and unstructured problems. It combines a database of internal and external data with a model base of analytical models and an interactive interface. Managers can carry out what-if analysis, goal seeking and forecasting to compare options. Unlike an MIS, which produces fixed reports, a DSS is flexible and interactive, but it supports rather than replaces human judgement.

Expert systems

An expert system captures the knowledge of human experts in a specific field. Its knowledge base holds facts and IF–THEN rules; its inference engine applies the rules to the facts supplied, using forward or backward chaining, to reach conclusions; its user interface asks questions and gives advice, often with an explanation facility. Expert systems give consistent, always-available advice but cannot use common sense, are costly to build and maintain, and raise questions of responsibility when wrong.

T Key terms from the mark schemes

Transaction processing system (TPS)
An information system that records and processes the routine day-to-day transactions of an organisation, in real time or in batches.

Management information system (MIS)
An information system that summarises data from transaction systems into regular, exception and on-demand reports for middle managers.

Decision support system (DSS)
An interactive information system that uses data and models to help managers make semi-structured and unstructured decisions, for example through what-if analysis.

Expert system
A computer system that emulates the decision-making of a human expert in a specific domain using a knowledge base and an inference engine.

Knowledge base
The part of an expert system that stores facts and rules gathered from human experts.

Inference engine
The part of an expert system that applies the rules in the knowledge base to the facts supplied in order to reach conclusions.

Exam focus

Questions often ask you to describe a system and apply it to a scenario, so state its purpose, its users and an example of its inputs and outputs in context. For expert systems, name all three components and say what each does – listing names alone rarely gains full marks. In discussion questions about expert systems, balance consistency and availability against the lack of common sense, cost and accountability, and conclude whether the system should support or replace the expert.

? Check yourself

1. State the difference between batch and real-time transaction processing, giving an example of each. [4]
2. Describe three types of report that a management information system might produce. [3]
3. Explain how a decision support system differs from a management information system. [3]

Answers: 1. Batch processing: transactions are collected together and processed as a group later, e.g. at night (1) / example: payroll or utility billing (1) / Real-time processing: each transaction is processed immediately and records are updated straight away (1) / example: cinema seat booking or bank withdrawal at an ATM (1). 2. Scheduled (periodic) reports produced at regular intervals, e.g. weekly sales by branch (1) / exception reports produced only when something falls outside normal limits, e.g. stores below target or hotel occupancy under 60% (1) / on-demand (ad hoc) reports produced when a manager requests them, e.g. a query on a specific product (1). Accept summary/detailed reports with example. 3. An MIS produces regular, structured reports, mostly from internal historical data, to support structured/semi-structured tactical decisions (1) / a DSS supports semi-structured or unstructured (often strategic) decisions using internal and external data and a model base (1) / it is interactive, allowing what-if analysis, goal seeking and forecasting, e.g. 'what happens to profit if fuel prices rise 10%?' (1).

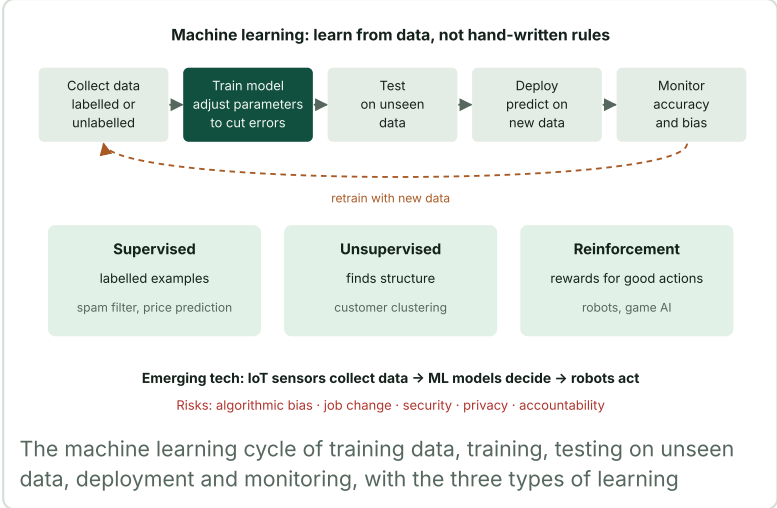
Artificial intelligence, machine learning and emerging technologies

IS16 FACT FILE

Specification: students should be able to

- Describe artificial intelligence and machine learning, including supervised, unsupervised and reinforcement learning
- Describe emerging technologies including the Internet of Things and robotics
- Evaluate the impact of these technologies on organisations

Key words: artificial intelligence, machine learning, training data, supervised learning, unsupervised learning, reinforcement learning, neural network, algorithmic bias, Internet of Things, sensor, robotics, automation



The machine learning cycle of training data, training, testing on unseen data, deployment and monitoring, with the three types of learning

AI and machine learning

Artificial intelligence is the ability of computer systems to perform tasks that normally require human intelligence. Machine learning is a branch of AI in which algorithms learn from data rather than following explicitly programmed rules. A model is trained on training data, tested on unseen data, then deployed to make predictions. Supervised learning uses labelled data, unsupervised learning finds patterns in unlabelled data, and reinforcement learning learns by trial and error using rewards.

Emerging technologies

The Internet of Things connects physical objects such as meters, vehicles and medical monitors so they can collect and exchange data automatically. Robotics uses programmable machines to sense and act in the physical world, from factory assembly to warehouse picking and surgery. Combined with AI, these technologies enable automation, predictive maintenance and real-time decision-making across many sectors, including agriculture, health, transport and retail.

Impact on organisations

AI and emerging technologies can increase productivity, reduce costs and errors, support better decisions and create new services. However, they can displace workers, require costly investment and new skills, and introduce risks: biased or unexplainable decisions, security weaknesses in IoT devices, and privacy concerns from large-scale data collection. Organisations must comply with data protection law, keep humans responsible for important decisions, and plan for staff retraining.

T Key terms from the mark schemes

Artificial intelligence (AI)
The ability of computer systems to perform tasks that normally require human intelligence, such as recognising images, understanding language and making decisions.

Machine learning
A branch of AI in which algorithms improve their performance at a task by learning patterns from data rather than following explicitly programmed rules.

Supervised learning
Machine learning in which a model is trained on labelled examples whose correct outputs are known.

Algorithmic bias
Systematic unfairness in the outputs of an AI system, usually caused by biased or unrepresentative training data.

Internet of Things (IoT)
A network of everyday physical objects with embedded sensors, processors and connectivity that collect and exchange data.

Exam focus

Extended answers on AI and emerging technologies reward depth and application. Define the technology precisely, give specific uses in the organisation named, and weigh benefits against drawbacks for different stakeholders – the organisation, employees, customers and society. Avoid science-fiction claims; current AI is narrow. Use technical terms such as training data, algorithmic bias and automation correctly, and finish with a justified conclusion rather than a summary.

? Check yourself

1. Define machine learning. [2]
2. Distinguish between supervised and unsupervised learning, giving an example of each. [4]
3. Explain what is meant by algorithmic bias and how it can occur. [2]

Answers: 1. Machine learning is a type of AI in which algorithms improve their performance at a task by learning from data (1) / rather than being explicitly programmed with rules – the model finds patterns in training data and uses them to make predictions on new data (1). 2. Supervised learning trains the model on labelled data, i.e. inputs with the correct outputs (1) / example: emails labelled spam/not spam to classify new emails; predicting house prices (1) / Unsupervised learning finds patterns or structure in unlabelled data with no correct answers given (1) / example: clustering customers into segments for marketing (1). 3. Algorithmic bias is when an AI system makes unfair or discriminatory decisions that systematically favour or disadvantage particular groups (1) / it can occur when the model is trained on biased or unrepresentative data, e.g. a recruitment model trained on past hiring that favoured men (1). Accept: biased choice of features or design by developers.

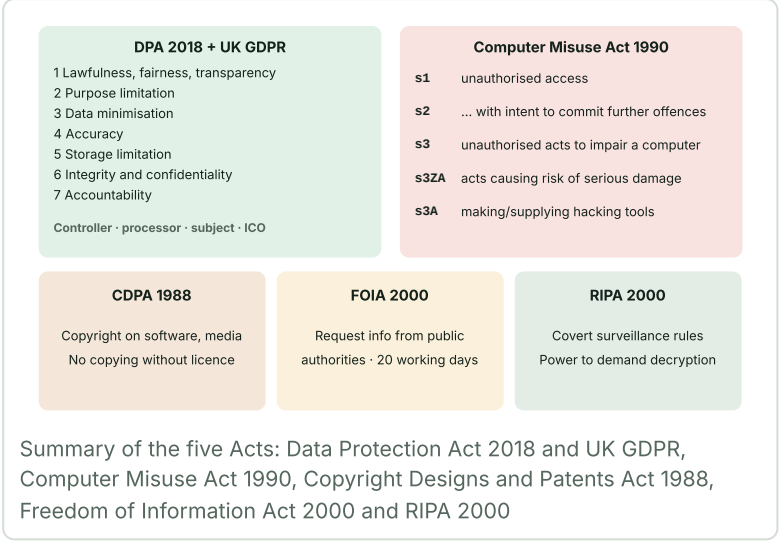
Legislation: data protection, computer misuse, copyright, FOI and RIPA

IS17 FACT FILE

Specification: students should be able to

- Describe the principles, rights and roles in the Data Protection Act 2018 and UK GDPR
- Describe the offences under the Computer Misuse Act 1990
- Describe the purpose of the Copyright, Designs and Patents Act 1988, the Freedom of Information Act 2000 and the Regulation of Investigatory Powers Act 2000
- Apply legislation to scenarios involving information systems

Key words: data subject, data controller, data processor, personal data, special category data, lawful basis, Information Commission (formerly the ICO), unauthorised access, copyright, software licence, freedom of information, interception



Summary of the five Acts: Data Protection Act 2018 and UK GDPR, Computer Misuse Act 1990, Copyright Designs and Patents Act 1988, Freedom of Information Act 2000 and RIPA 2000

Data protection

The UK GDPR and the Data Protection Act 2018 govern the processing of personal data about living individuals. Data controllers must follow seven principles: lawfulness, fairness and transparency; purpose limitation; data minimisation; accuracy; storage limitation; integrity and confidentiality; and accountability. Data subjects have rights including being informed, access, rectification, erasure, restriction, portability, objection and rights about automated decisions. The ICO regulates compliance. The Data (Use and Access) Act 2025 has amended parts of the law, with changes phased in.

Computer Misuse Act 1990

The Computer Misuse Act creates offences of unauthorised access to computer material (section 1), unauthorised access with intent to commit or facilitate further offences (section 2), unauthorised acts intended to impair the operation of a computer, such as spreading malware or denial-of-service attacks (section 3), unauthorised acts causing or risking serious damage (section 3ZA) and making, supplying or obtaining tools for these offences (section 3A). Penalties increase with seriousness, up to life imprisonment under section 3ZA.

Copyright, FOI and RIPA

The Copyright, Designs and Patents Act 1988 protects creators' works, including software, from being copied, adapted or distributed without permission; software is used under licence. The Freedom of Information Act 2000 gives the public the right to request information held by public authorities, normally answered within 20 working days, subject to exemptions. The Regulation of Investigatory Powers Act 2000 regulates covert surveillance and the power to require encrypted data to be decrypted, balancing investigation against privacy.

T Key terms from the mark schemes

Personal data
Any information relating to an identified or identifiable living individual.

Data controller
The person or organisation that decides the purposes for which, and the way in which, personal data is processed.

Data processor
A person or organisation that processes personal data on behalf of a data controller.

Special category data
Personal data needing extra protection, such as health, ethnicity, religion, political opinions, trade union membership, genetic and biometric data.

Freedom of Information Act 2000
UK law giving the public the right to request recorded information held by public authorities, subject to exemptions.

Data subject
The living individual whom personal data is about.

Computer Misuse Act 1990
UK law creating offences of unauthorised access to computer material, access with intent to commit further offences, and unauthorised acts that impair computers.

Exam focus

Name the precise Act and the relevant principle, right or section, then apply it to the scenario: 'the school has breached integrity and confidentiality because it failed to secure pupils' health data, which is special category data'. Avoid inventing principles such as 'data must be kept secure for six months'. Do not confuse the Freedom of Information Act (public authorities' information) with a subject access request under data protection law (your own personal data). Learn the roles: controller, processor, subject, ICO.

? Check yourself

1. State four of the seven data protection principles. [4]
2. Explain the difference between a data controller and a data processor. [2]
3. A pupil uses a teacher's password to log in and change their exam marks. Identify which offences under the Computer Misuse Act 1990 have been committed and explain why. [3]

Answers: 1. Any four from: lawfulness, fairness and transparency (1) / purpose limitation (1) / data minimisation (1) / accuracy (1) / storage limitation (1) / integrity and confidentiality (security) (1) / accountability (1). 2. A data controller decides why and how personal data is processed (the purposes and means) and is responsible for compliance, e.g. a school (1) / a data processor processes personal data on behalf of the controller, following its instructions, e.g. a cloud provider or payroll company (1). 3. Section 1: unauthorised access to computer material – the pupil logged in with a password they were not entitled to use (1) / Section 2: unauthorised access with intent to commit or facilitate a further offence – they accessed the system intending to change their marks, i.e. fraud/deception (1) / Section 3: unauthorised acts with intent to impair the operation of a computer / the reliability of data – they altered the exam marks stored on the system (1).

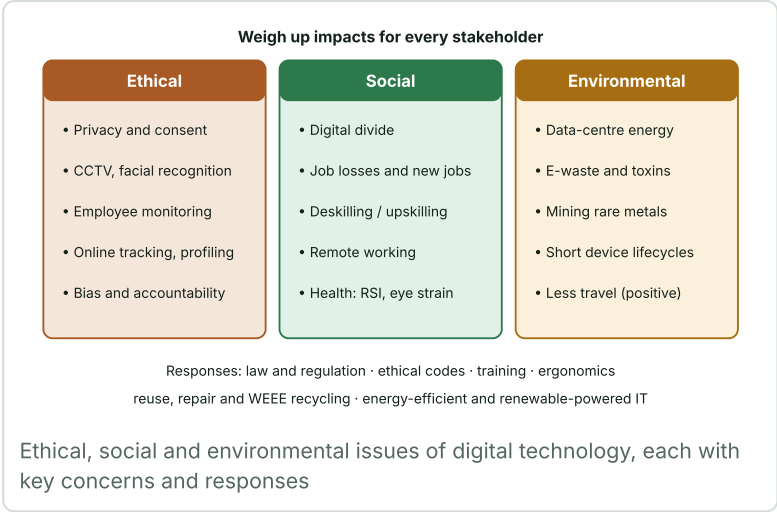
Ethical, social and environmental issues

IS18 FACT FILE

Specification: students should be able to

- Discuss ethical issues including privacy and surveillance
- Discuss social issues including the digital divide, employment and health and safety
- Discuss environmental issues including energy use, sustainability and e-waste

Key words: ethics, privacy, surveillance, digital divide, deskilling, upskilling, remote working, ergonomics, repetitive strain injury, sustainability, e-waste, carbon footprint



Ethical, social and environmental issues of digital technology, each with key concerns and responses

Privacy and surveillance

Digital technology makes it easy to collect, combine and analyse information about people through CCTV, facial recognition, location tracking, online cookies and employee monitoring. Supporters argue this improves security, crime prevention and convenience. Critics point to loss of privacy, chilling effects on free expression, the risk of misuse or data breaches, and inaccurate or biased systems. Surveillance should be lawful, necessary and proportionate, and people should be told what is collected and why.

Social issues

The digital divide separates people who have affordable access to devices, connectivity and digital skills from those who do not, excluding them from services, education and work. Automation removes some routine jobs while creating skilled ones, so retraining is essential. Remote working brings flexibility but can cause isolation. Health risks of prolonged computer use include repetitive strain injury, eye strain, back problems and stress, which are reduced by ergonomic equipment, breaks and sensible policies.

Environmental issues

Producing, powering and disposing of digital technology has environmental costs: mining of scarce materials, the large energy use of data centres and networks, and e-waste containing toxic substances. Short replacement cycles worsen the problem. Organisations can reduce their impact by extending device life, refurbishing and recycling through approved schemes, buying energy-efficient equipment, virtualising servers and using renewable-powered cloud providers. Technology can also help the environment through reduced travel and smarter energy use.

T Key terms from the mark schemes

Digital divide
The gap between people who have access to digital technology and the skills to use it, and those who do not.

Surveillance
The monitoring of people's behaviour, activities or communications, for example by CCTV, tracking or interception.

Sustainability
Meeting present needs without compromising the ability of future generations to meet theirs, for example by reducing energy use and e-waste.

E-waste
Discarded electrical and electronic devices, which may contain toxic and valuable materials.

Repetitive strain injury (RSI)
Pain or damage to muscles, tendons or nerves caused by repeated movements such as typing or using a mouse.

Exam focus

Ethical, social and environmental issues are often tested with extended QWC questions worth 10–15 marks. Plan quickly: identify several distinct issues, explain each with a technical point and an applied example, consider different stakeholders, and weigh positive against negative effects. Use paragraphs and technical vocabulary. A concluding judgement that follows from your arguments is needed for the top band; a list of unexplained points stays in the lowest band however long it is.

? Check yourself

1. Define the term digital divide. [2]
2. Give two causes of the digital divide in Northern Ireland and suggest a way to reduce each. [4]
3. Describe two health problems that can result from prolonged computer use and how each can be prevented. [4]

Answers: 1. The digital divide is the gap between people who have access to digital technology (devices and internet) (1) / and the skills/confidence to use it, and those who do not (1). 2. Two causes (1 each) with a matching solution (1 each), e.g.: poor rural broadband connectivity, e.g. in parts of Fermanagh (1) / government broadband roll-out schemes (e.g. Project Stratum) or satellite/4G-5G alternatives (1) / cost of devices and broadband for low-income households (1) / device lending schemes, subsidised social tariffs (1) / lack of skills or confidence, especially among older people (1) / free digital-skills classes in libraries and community centres (1) / disability (1) / accessible design and assistive technology (1). 3. Two problems (1 each) with prevention (1 each), e.g.: repetitive strain injury (RSI) from repeated keyboard/mouse use (1) / ergonomic keyboards, wrist rests and regular breaks (1) / eye strain and headaches from long periods at screens and glare (1) / anti-glare screens, correct lighting, breaks, free eye tests (1) / back and neck problems from poor posture or unsuitable chairs (1) / adjustable chairs and screens, posture training (1) / stress (1) / workload management, right-to-disconnect policies (1). Accept reference to the Health and Safety at Work (NI) Order 1978 / display screen equipment regulations.